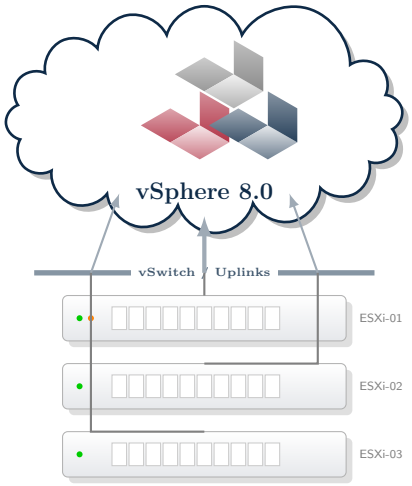
Information Technology Institute

Alexandria Branch



VMware vSphere

Infrastructure Implementation Project

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System Administration Track — Intake 46

2026

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1 Introduction

VMware vSphere is a virtualization platform that allows multiple virtual machines to run on physical servers. It helps organizations reduce hardware costs, improve efficiency, and manage IT resources more effectively.
vSphere provides tools for creating, managing, and monitoring virtual machines in a data center environment. The platform includes ESXi (the hypervisor) and vCenter Server (the management tool).
This project documents the implementation of a vSphere environment. We set up an Active Directory Domain Controller for centralized authentication, installed ESXi hosts and joined them to the domain, deployed vCenter Server and integrated it with Active Directory, set up virtual networks and storage, and deployed virtual machines. This hands-on project demonstrates the essential skills needed to build and manage a virtualization infrastructure integrated with enterprise authentication services.

1.1 Project Objectives

The main goals of this project are:

1. **Set Up Active Directory Domain Controller**
   - Deploy Windows Server with iti.local domain
   - Configure DNS and create user accounts for vSphere management
2. **Install and Configure ESXi Hosts**
   - Install VMware ESXi on three host servers
   - Join ESXi hosts to iti.local domain and verify connectivity
3. **Deploy vCenter Server**
   - Install vCenter Server and integrate with Active Directory
   - Create datacenter and cluster, then add ESXi hosts
4. **Configure Virtual Networking and Storage**
   - Create standard virtual switches for network connectivity
   - Configure shared storage and create datastores for VMs
5. **Deploy and Manage Virtual Machines**
   - Create VMs with different operating systems
   - Install VMware Tools and configure VM resources
6. **Implement Resource Management**
   - Enable DRS for automatic load balancing across hosts
   - Configure resource limits and reservations for VMs
7. **Enable High Availability and Fault Tolerance**
   - Configure vSphere HA for automatic VM restart on host failure
   - Implement Fault Tolerance for critical VMs requiring zero downtime

1.2 Network Topology

Figure 1 illustrates the complete vSphere infrastructure implemented in this project.

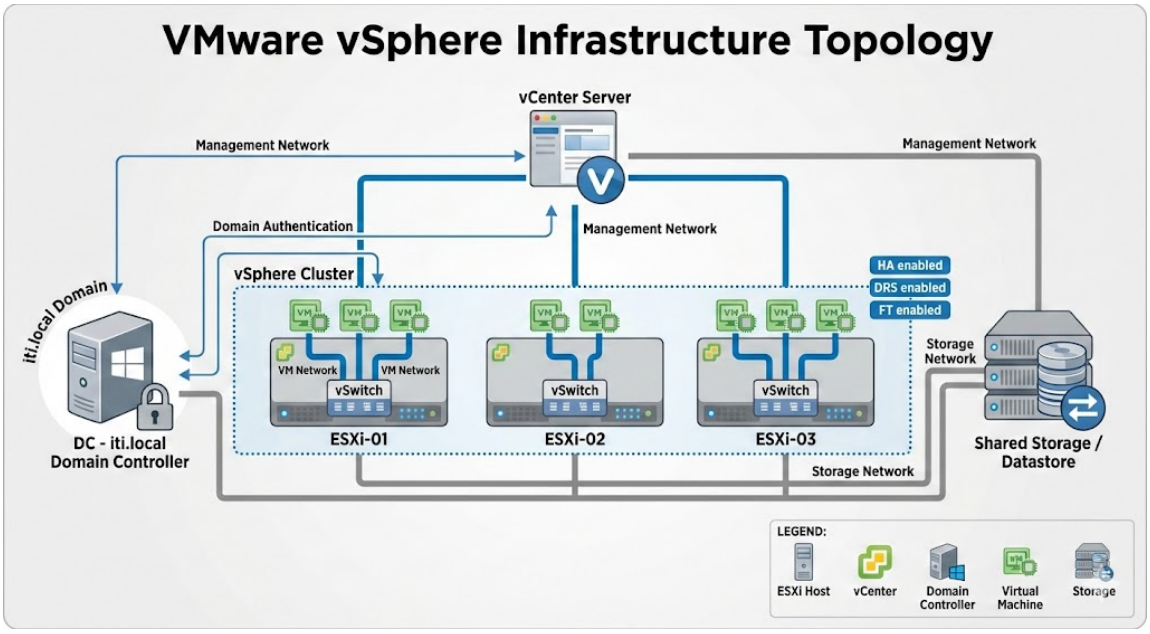

Figure 1: Network Topology Diagram showing vCenter Server, ESXi hosts cluster with HA/DRS/FT enabled, Domain Controller (iti.local), and NFS shared storage

1.3 Network Configuration Summary

The following tables detail the IP addressing scheme and network configuration for all infrastructure components.

1.3.1 Network Parameters

Parameter	Value
Network Address	10.153.153.0
Subnet Mask	255.255.255.0
Primary DNS Server	10.153.153.1 (DC1)
Domain Name	iti.local

Table 1: Network Configuration Parameters

1.3.2 IP Address Assignment

Device	IP Address	Role/Description
DC1	10.153.153.1	Domain Controller (iti.local) + DNS
ESXi-Hamdy	10.153.153.33	ESXi Host
ESXi-Jimmy	10.153.153.113	ESXi Host
ESXi-Omar	10.153.153.232	ESXi Host
vCenter Server	10.153.153.34	vCenter Server (Centralized Management)
NFS-Storage	10.153.153.153	Linux NFS Shared Storage Server

Table 2: Infrastructure IP Address Allocation

1.3.3 Virtual Switch Configuration

ESXi Host	vSwitch IP Address	Purpose
ESXi-Hamdy	10.153.153.80	vMotion
ESXi-Jimmy	10.153.153.82	vMotion
ESXi-Omar	10.153.153.81	vMotion

Table 3: Virtual Switch Assignment per ESXi Host

2 Active Directory Integration

2.1 Domain Controller Setup

A Windows Server was deployed and configured as the Active Directory Domain Controller for the iti.local domain. DNS services were configured to support domain name resolution for all infrastructure components.

2.2 ESXi Hosts Domain Join

2.2.1 Configuring ESXi DNS Settings

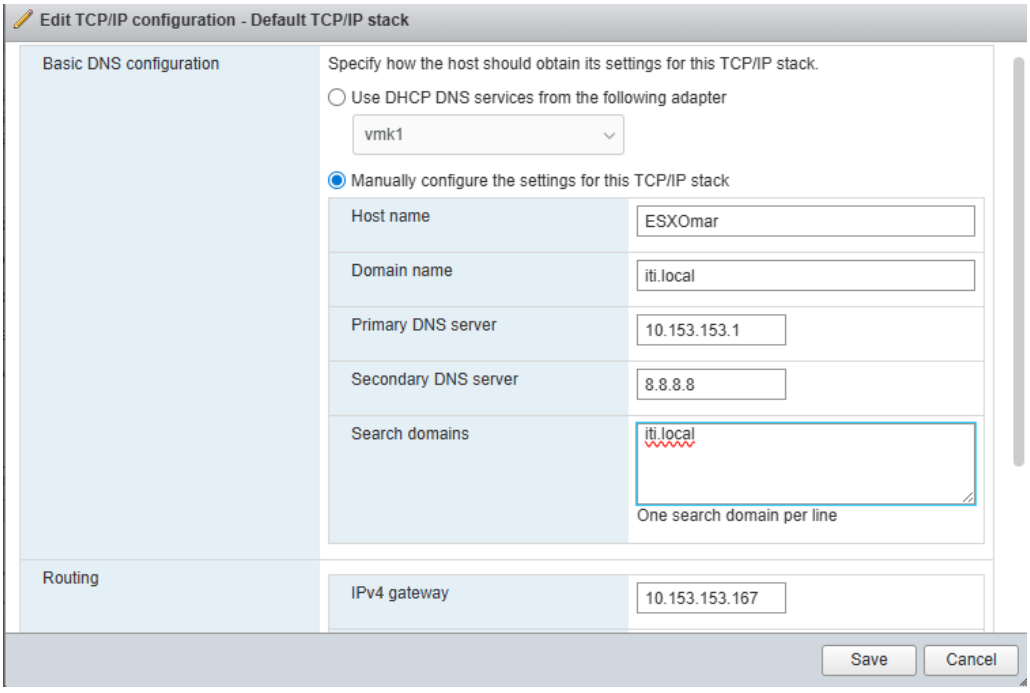

Figure 2: ESXi TCP/IP configuration for FQDN setup, including hostname, domain name (iti.local), and primary DNS server (10.153.153.1) to enable full domain integration

2.2.2 Joining ESXi to iti.local Domain

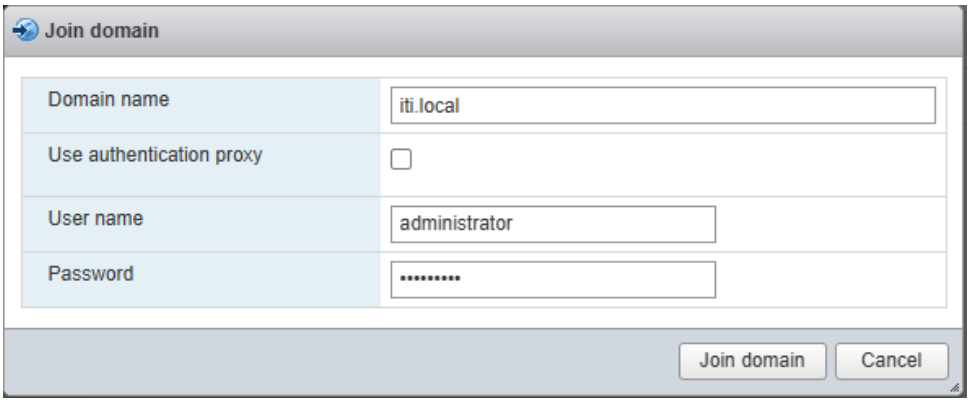

Figure 3: ESXi host joining iti.local domain using administrator credentials

2.3 vCenter Server Domain Join

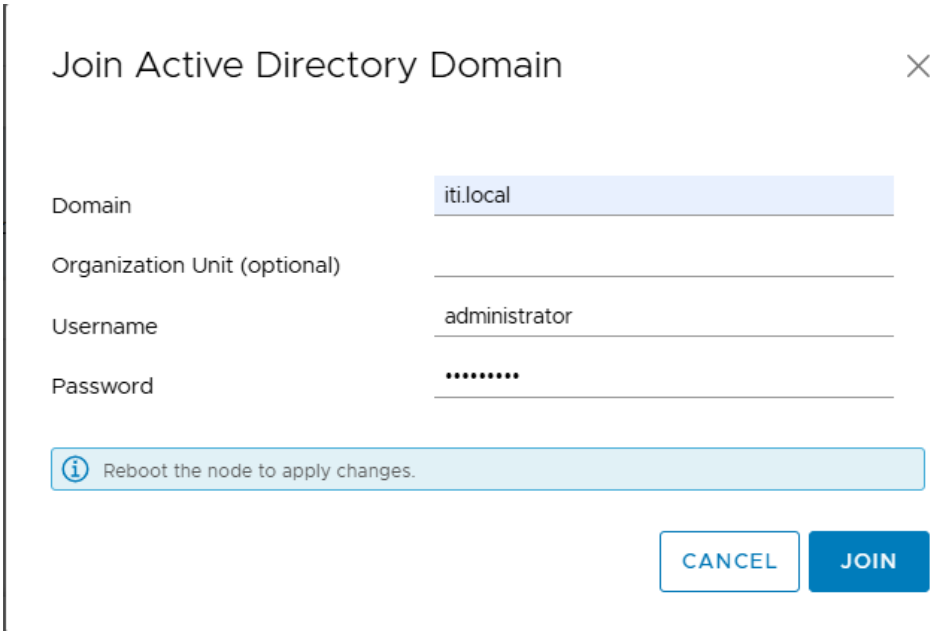

Figure 4: vCenter Server joining iti.local Active Directory domain with administrator authentication via vCenter Management Interface (Port 5480)

2.4 Linux NFS Server Domain Join

2.4.1 Installing Required Packages

```
[root@ahmed ~]# sudo yum install realmd sssd oddjob oddjob-mkhomedir adcli sambe-common-tools -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:44:39 ago on Sun 15 Feb 2026 10:20:41 AM EST.
Package realmd-0.17.1-2.el9.x86_64 is already installed.
Package sssd-2.9.4-2.el9.x86_64 is already installed.
Package adcli-0.9.2-1.el9.x86_64 is already installed.
No match for argument: sambe-common-tools
Error: Unable to find a match: sambe-common-tools
[root@ahmed ~]# sudo yum install realmd sssd oddjob oddjob-mkhomedir adcli samba-common-tools -y
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-manager" to register.

Last metadata expiration check: 0:45:07 ago on Sun 15 Feb 2026 10:20:41 AM EST.
Package realmd-0.17.1-2.el9.x86_64 is already installed.
Package sssd-2.9.4-2.el9.x86_64 is already installed.
Package adcli-0.9.2-1.el9.x86_64 is already installed.
Dependencies resolved.
=================================================================================================================
 Package                          Architecture        Version                     Repository                Size
=================================================================================================================
Installing:
 oddjob                           x86_64              0.34.7-7.el9                Local_AppStream            73 k
 oddjob-mkhomedir                 x86_64              0.34.7-7.el9                Local_AppStream            31 k
 samba-common-tools               x86_64              4.19.4-104.el9              Local_BaseOS              475 k
Installing dependencies:
 libnetapi                        x86_64              4.19.4-104.el9              Local_BaseOS              144 k
 samba-ldb-ldap-modules           x86_64              4.19.4-104.el9              Local_BaseOS               30 k
 samba-libs                       x86_64              4.19.4-104.el9              Local_BaseOS              130 k

Transaction Summary
```

Figure 5: Installing required packages (realmd, sssd, oddjob, oddjob-mkhomedir, adcli, samba-common-tools) on Linux NFS server for Active Directory integration

2.4.2 Joining Linux Server to iti.local Domain

```
[root@NFS-Server ~]# sudo realm join iti.local -U administrator
Password for administrator@ITI.LOCAL:
[root@NFS-Server ~]# sudo realm discover iti.local
iti.local
  type: kerberos
  realm-name: ITI.LOCAL
  domain-name: iti.local
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: oddjob
  required-package: oddjob-mkhomedir
  required-package: sssd
  required-package: adcli
  required-package: samba-common-tools
  login-formats: %U@iti.local
  login-policy: allow-realm-logins
[root@NFS-Server ~]#
```

Figure 6: Linux NFS server successfully joined to iti.local domain using realm command with administrator credentials, showing Kerberos authentication configuration

2.5 Active Directory Structure Verification

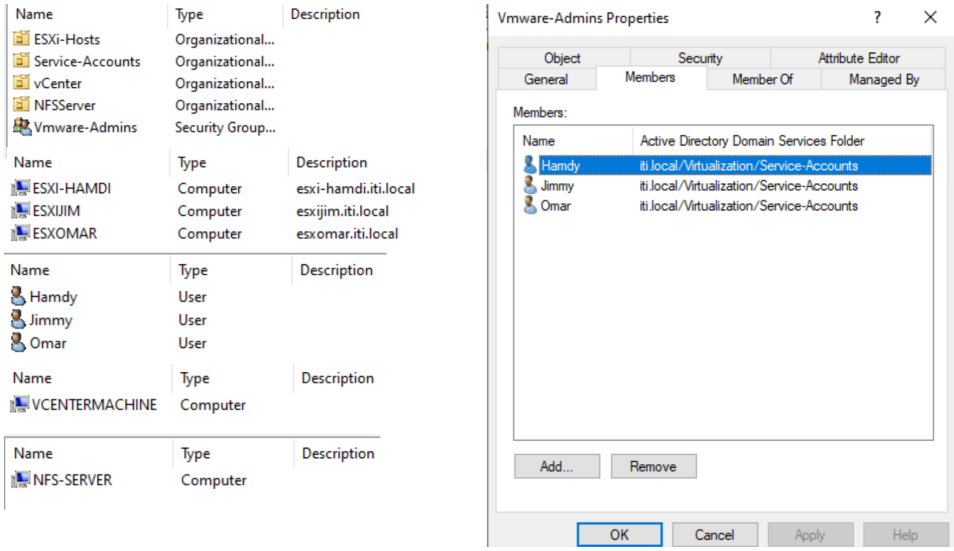

Figure 7: Active Directory organizational structure and Vmware-Admins security group configuration

The Active Directory structure shows the complete integration of all infrastructure components into the iti.local domain:

Left Panel - Organizational Structure:

- **Organizational Units (OUs):** ESXi-Hosts, Service-Accounts, vCenter, NFSServer
- **Domain-Joined Computers:** ESXI-HAMDI, ESXIJIM, ESXOMAR, VCENTERMACHINE, NFS-SERVER
- **User Accounts:** Hamdy, Jimmy, Omar

Right Panel - Vmware-Admins Security Group:

- **Group Members:** Hamdy, Jimmy, Omar
- **Purpose:** Administrative privileges for vSphere management

2.6 DNS Configuration

DNS services were configured on the Domain Controller to provide name resolution for all infrastructure components.

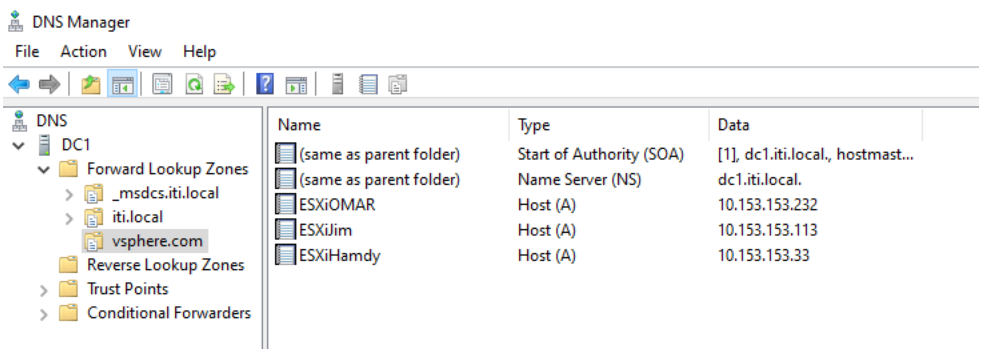

Figure 8: DNS Manager showing Forward Lookup Zones with iti.local domain entries for DC1, ESXi hosts (ESXiOMAR, ESXiJim, ESXiHamdy), and vsphere.com zone

2.7 ESXi Host Permissions Configuration

After joining the ESXi hosts to the iti.local domain, the Vmware-Admins security group was granted administrator access to enable domain users to manage the ESXi hosts.

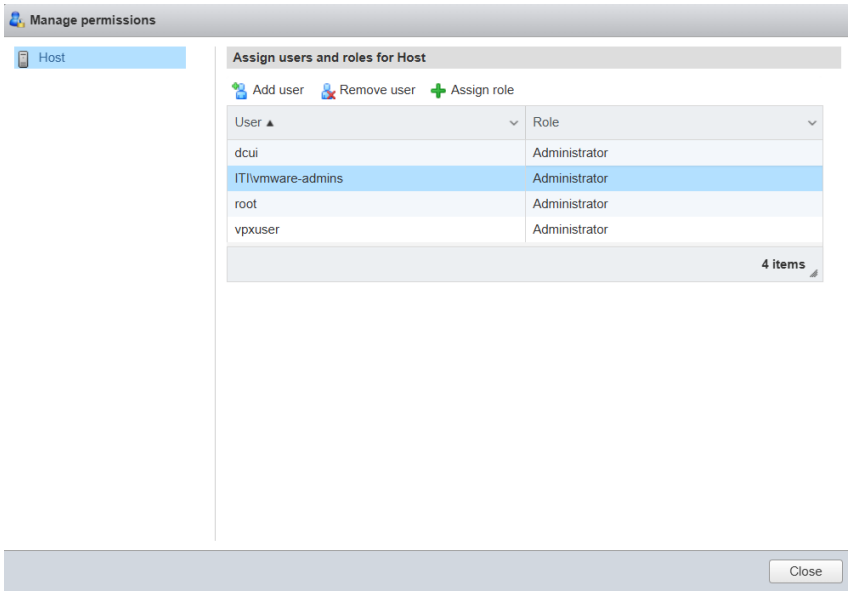

Figure 9: ESXi Host permissions showing ITI\vmware-admins group assigned with Administrator role

2.8 Domain Authentication Verification

To verify that the Active Directory integration was successful, login tests were performed using domain credentials.

2.8.1 Permission Denied for Non-Authorized Users

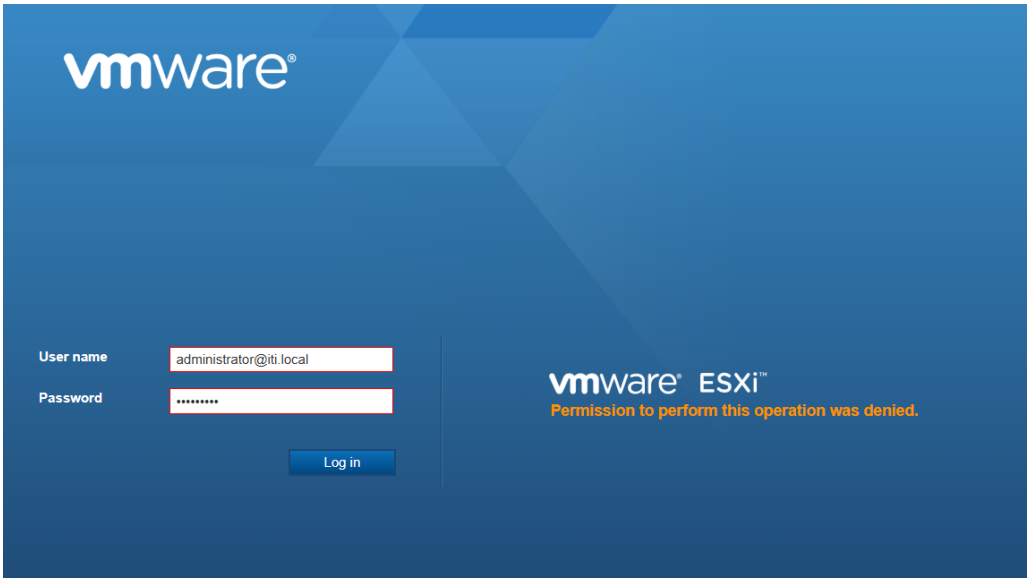

Figure 10: ESXi login attempt with administrator@iti.local showing "Permission to perform this operation was denied" - demonstrating proper access control

The administrator domain account was correctly denied access as it is not a member of the Vmware-Admins group.

2.8.2 Successful Login with Authorized Domain User

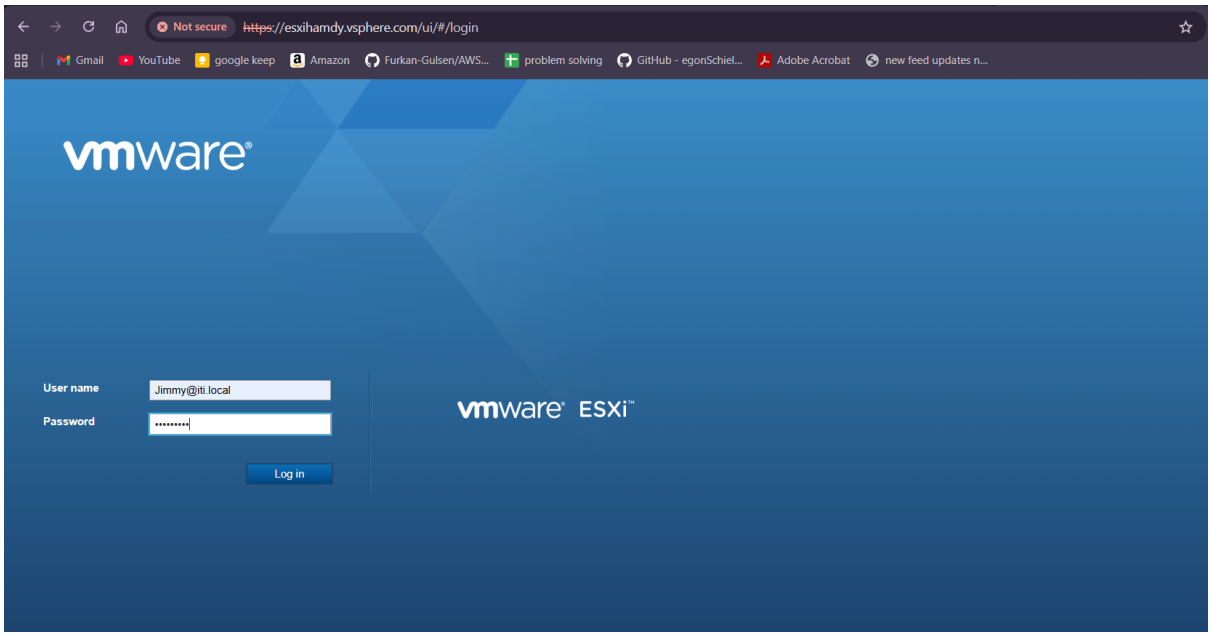

Figure 11: ESXi login screen with Jimmy@iti.local domain credentials

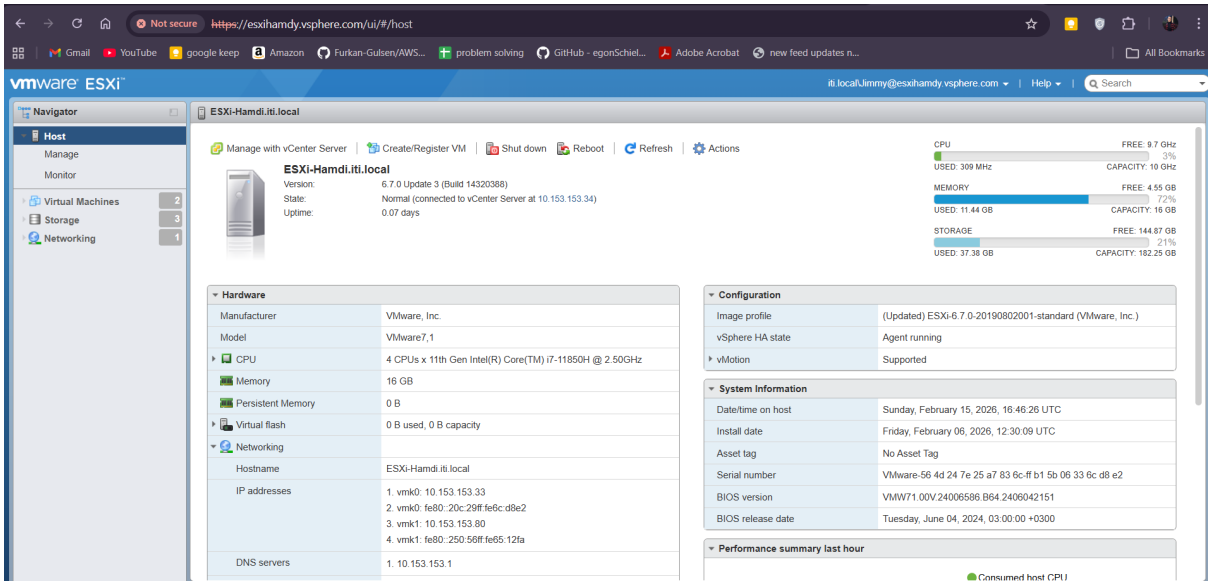

Figure 12: ESXi host dashboard after successful login showing authenticated user iti.local\Jimmy@esxihamdy.vsphere.com with full administrative access

3 vCenter Infrastructure Configuration

3.1 Creating Datacenter

A datacenter is the top-level container in the vCenter inventory that organizes compute resources.

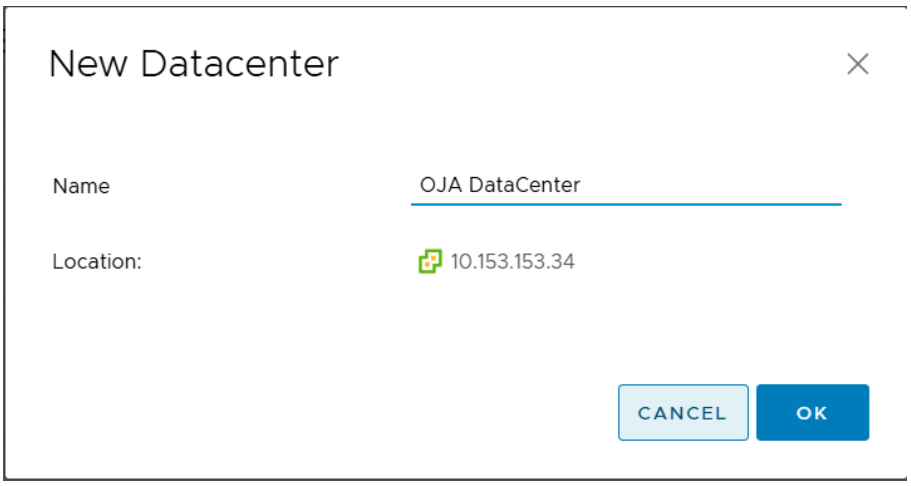

Figure 13: Creating new datacenter "OJA DataCenter" in vCenter Server at location 10.153.153.34

3.2 Creating Cluster

A cluster groups multiple ESXi hosts to enable resource pooling and high availability features.

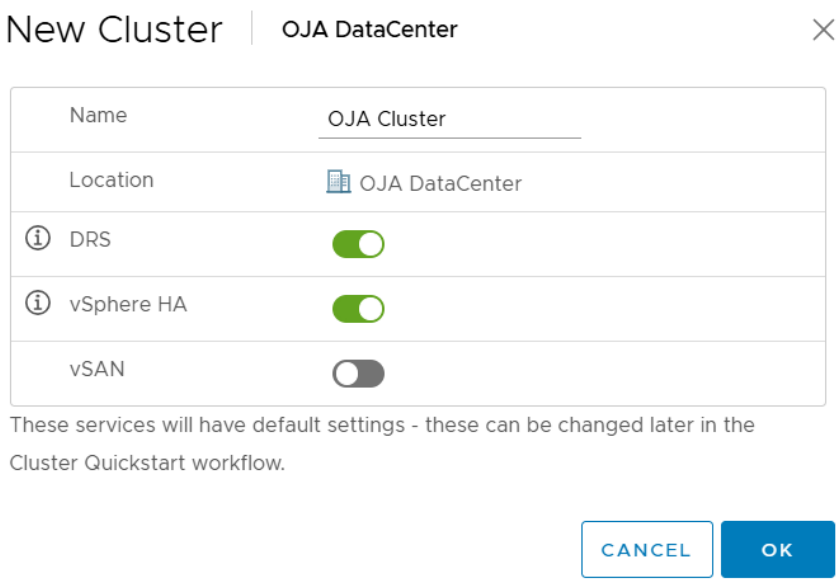

Figure 14: Creating "OJA Cluster" within OJA DataCenter with DRS and vSphere HA enabled, vSAN disabled

The cluster was configured with the following services:

- **DRS (Distributed Resource Scheduler):** Enabled - for automatic load balancing

- **vSphere HA (High Availability):** Enabled - for automatic VM restart on host failure
- **vSAN:** Disabled - using external NFS storage instead

3.3 Adding ESXi Hosts to Cluster

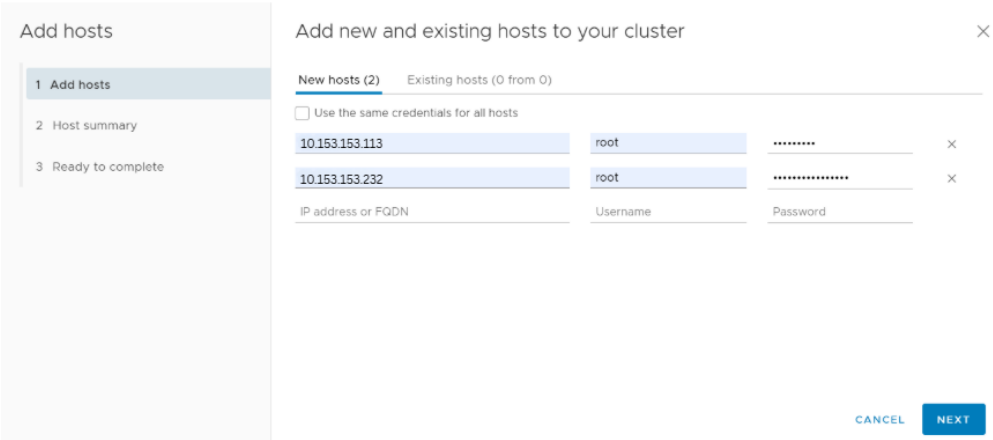

Figure 15: Adding ESXi hosts to OJA Cluster - ESXi-Jimmy (10.153.153.113) and ESXi-Omar (10.153.153.232) with root credentials

4 Content Library and vMotion Network Configuration

4.1 Creating Content Library

A content library provides a centralized repository for storing and managing VM templates, ISO images, and OVF packages.

4.1.1 Naming Content Library

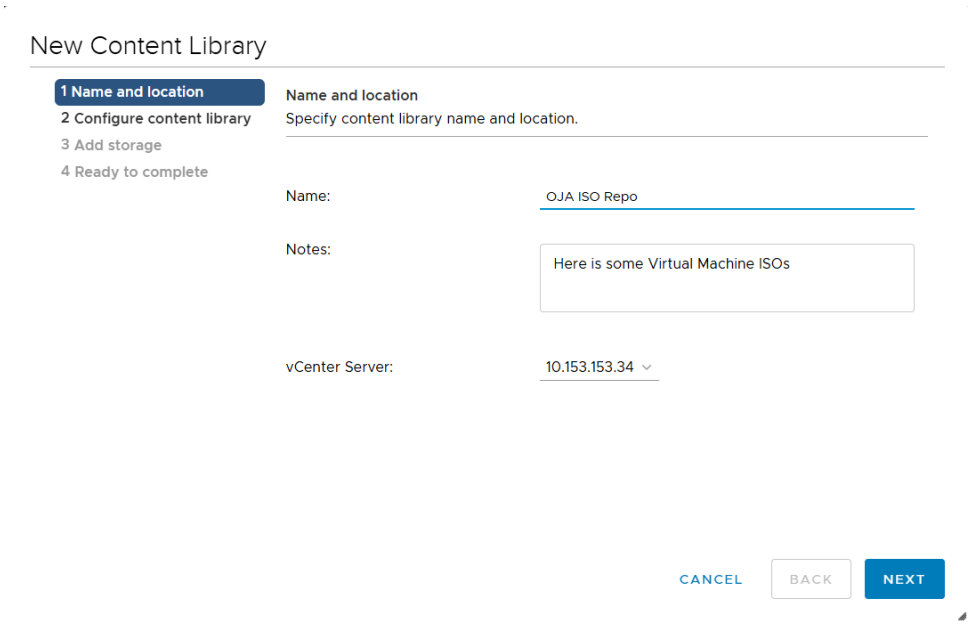

Figure 16: Creating new content library "OJA ISO Repo" for storing virtual machine ISOs on vCenter Server (10.153.153.34)

4.1.2 Configuring Library Type

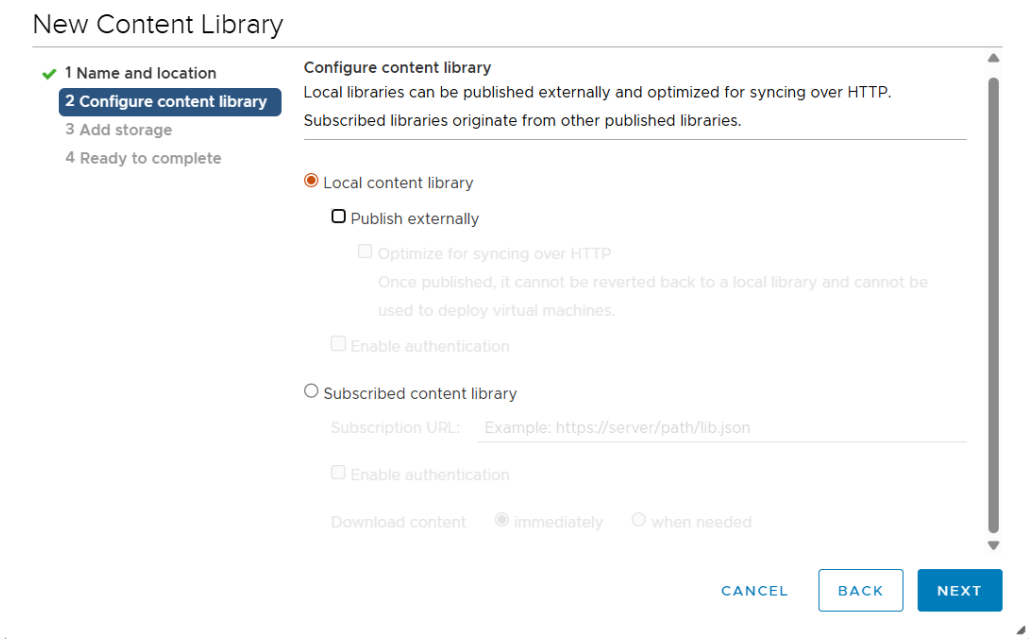

Figure 17: Configuring content library as local content library without external publishing

4.1.3 Selecting Storage Location

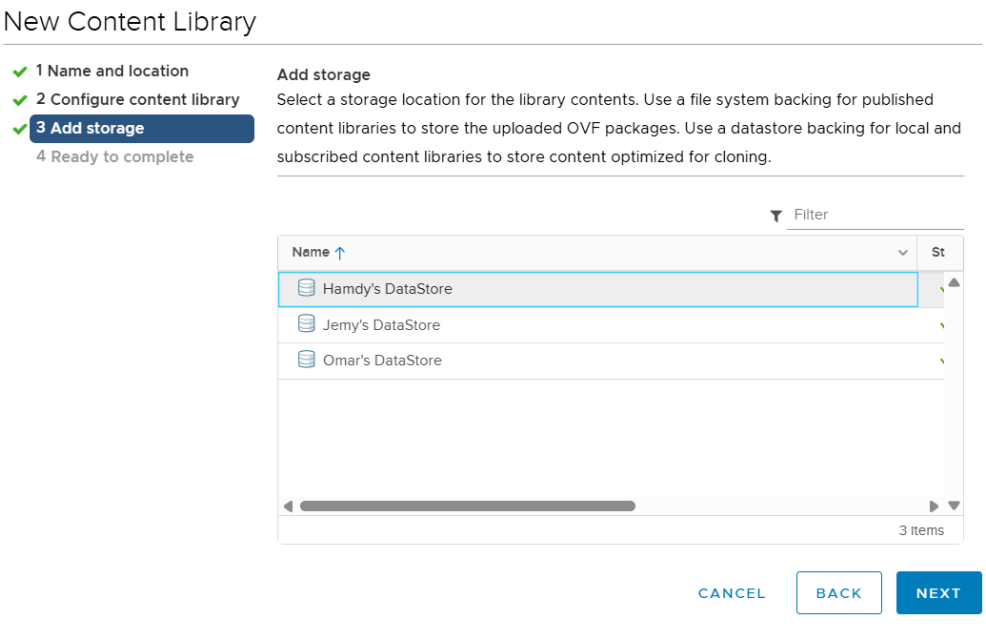

Figure 18: Selecting Hamdy's DataStore as storage location for content library contents

4.1.4 Content Library Verification

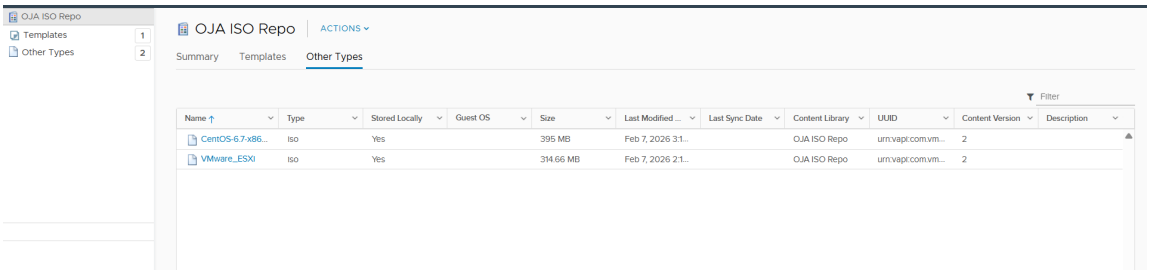

Figure 19: OJA ISO Repo content library showing uploaded ISO files including CentOS-6.7 (395 MB) and VMware_ESXi (314.66 MB)

4.2 Configuring vMotion Network

vMotion enables live migration of running virtual machines between ESXi hosts without downtime. A dedicated network with standard virtual switches was configured on each ESXi host for vMotion traffic.

4.2.1 ESXi-Hamdy vMotion Configuration

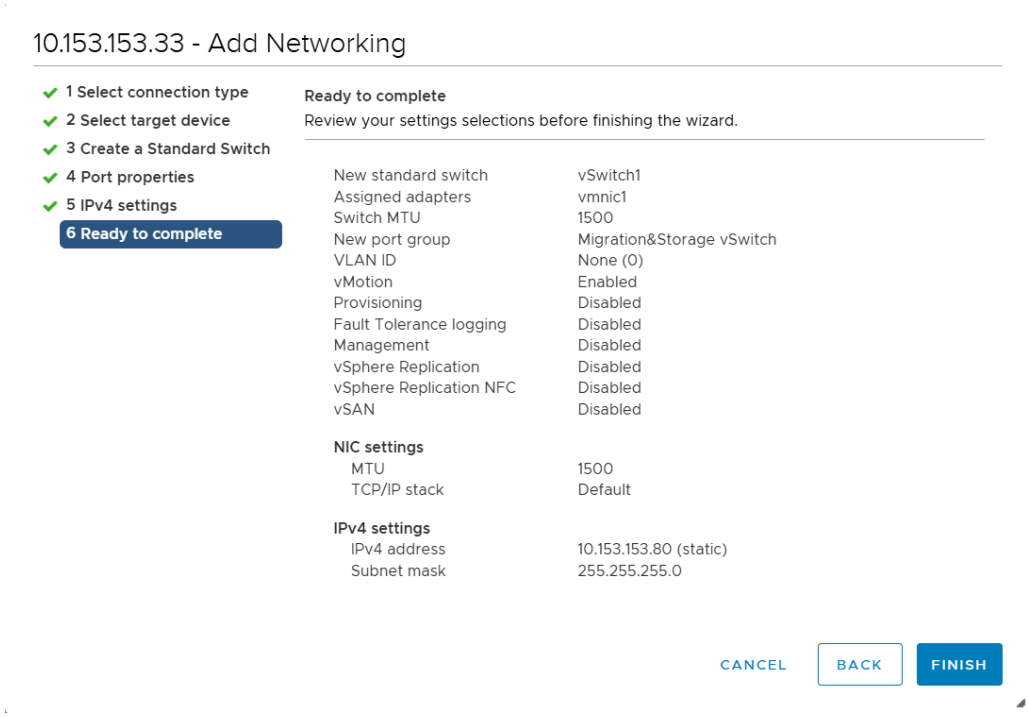

Figure 20: vSwitch1 configuration on ESXi-Hamdy (10.153.153.33) with vMotion enabled, port group "Migration&Storage vSwitch", and static IP 10.153.153.80

4.2.2 ESXi-Jimmy vMotion Configuration

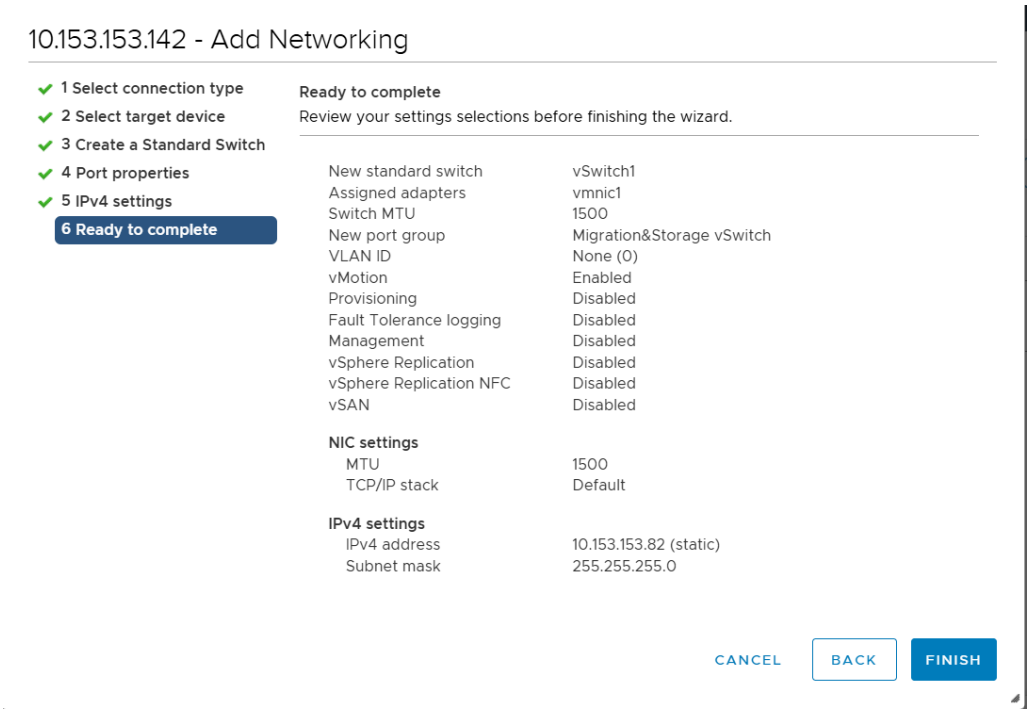

Figure 21: vSwitch1 configuration on ESXi-Jimmy (10.153.153.142) with vMotion enabled, port group "Migration&Storage vSwitch", and static IP 10.153.153.82

4.2.3 ESXi-Omar vMotion Configuration

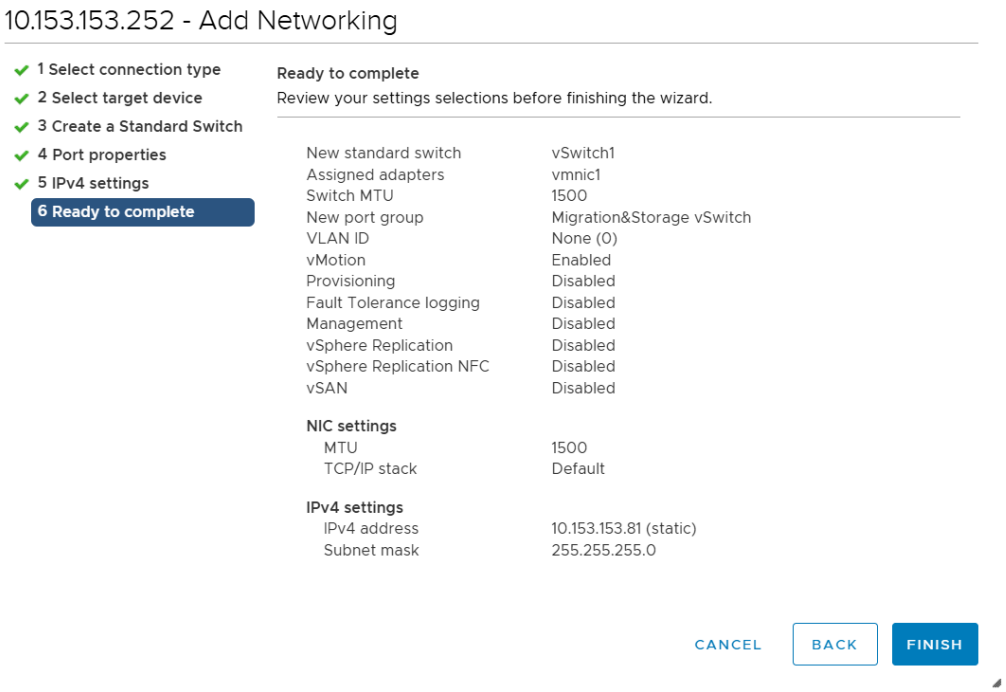

Figure 22: vSwitch1 configuration on ESXi-Omar (10.153.153.252) with vMotion enabled, port group "Migration&Storage vSwitch", and static IP 10.153.153.81

4.3 Virtual Machine Deployment from Content Library

After creating the content library and uploading ISO files, VM templates can be created and stored in the content library for rapid deployment.

4.3.1 Creating VM Template in Content Library

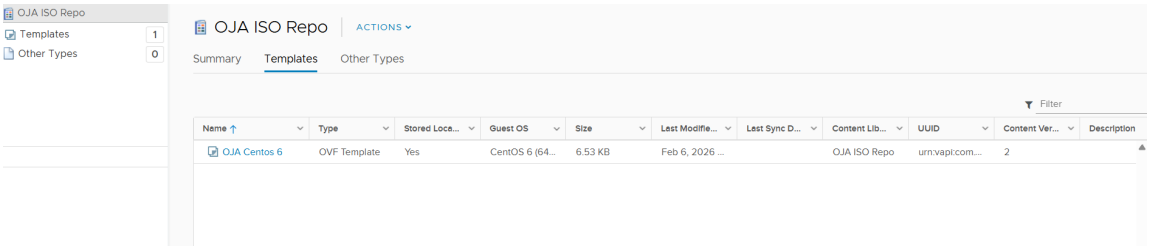

Figure 23: OJA ISO Repo content library Templates section showing "OJA Centos 6" OVF template (CentOS 6 64-bit, 6.53 KB) ready for deployment

4.3.2 Deploying VM from Template

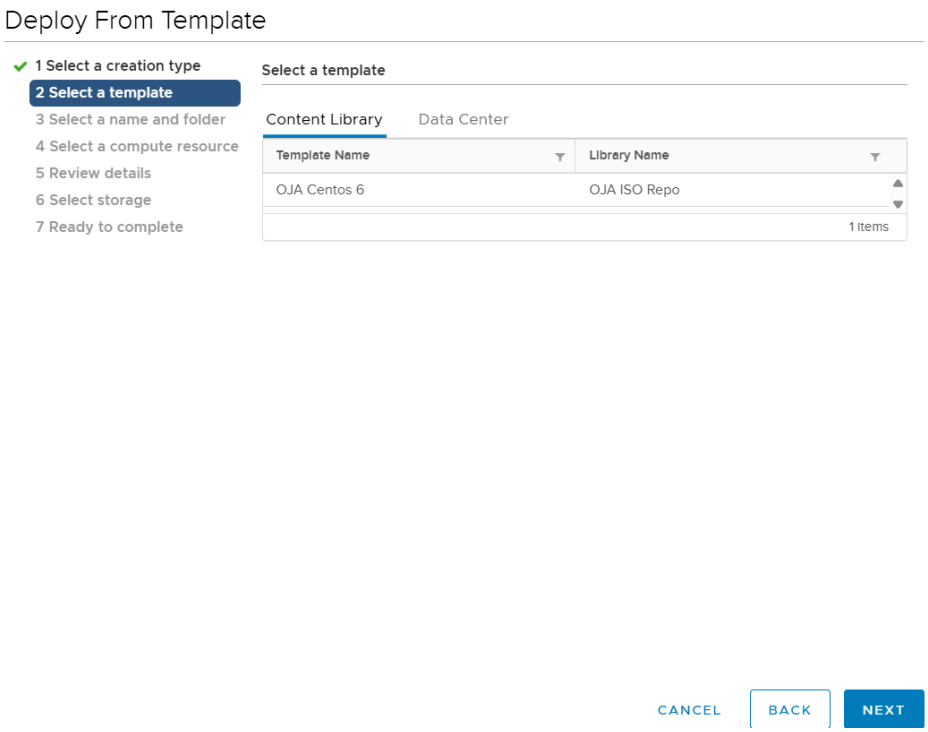

Figure 24: Selecting "OJA Centos 6" template from OJA ISO Repo content library for VM deployment

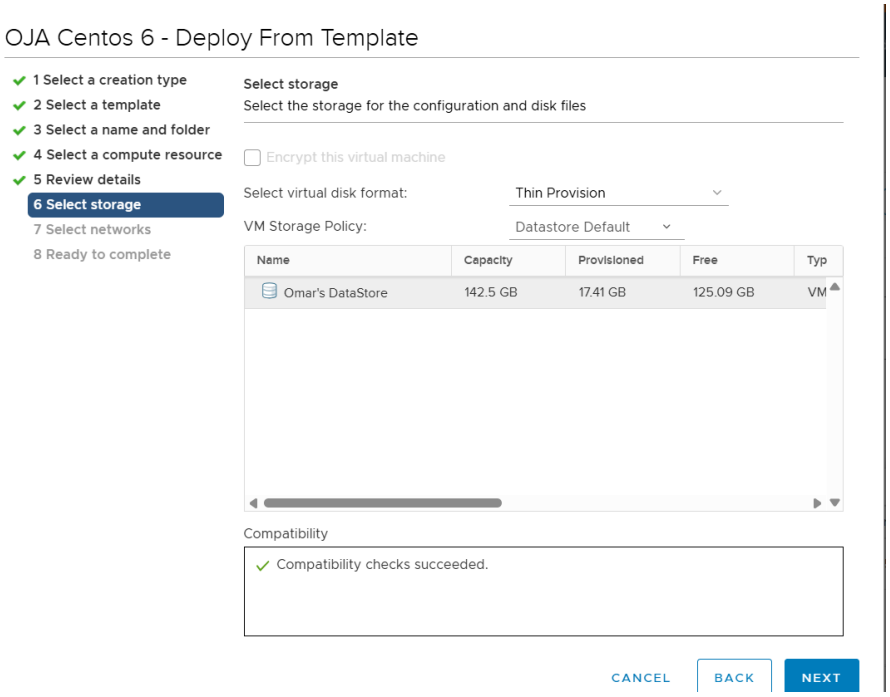

Figure 25: Selecting Omar's DataStore (142.5 GB capacity, 125.09 GB free) with Thin Provision disk format for VM deployment, compatibility checks succeeded

5 NFS Shared Storage Configuration

5.1 NFS Server Setup

A dedicated Linux server was configured to provide centralized NFS This shared storage enables features like vMotion, High Availability, and Distributed Resource Scheduler.

5.1.1 Installing NFS Utilities and Service Status

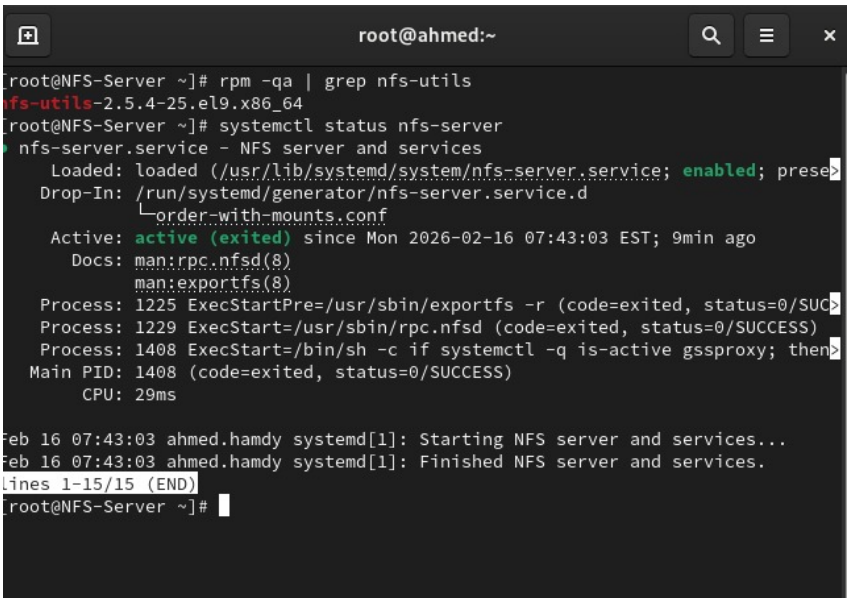

Figure 26: NFS utilities package (nfs-utils-1-2.5.4-25) installed and nfs-server service enabled and active on Linux server

5.1.2 Creating and Mounting Storage Partition

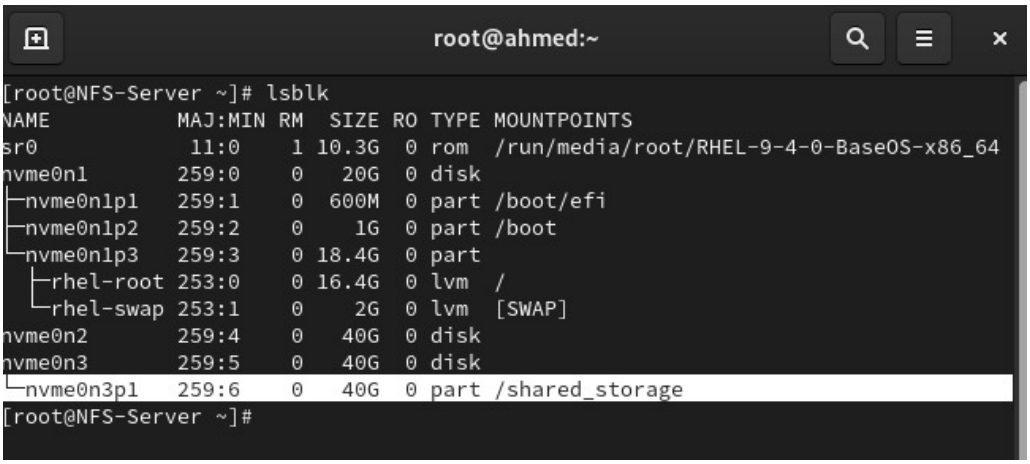

Figure 27: Disk partition layout showing /shared_storage mounted on nvme0n3p1 (40GB partition) for NFS exports

A dedicated 40GB partition was created and mounted at `/shared_storage` to serve as the NFS export directory.

5.1.3 NFS Export Configuration and Firewall Setup

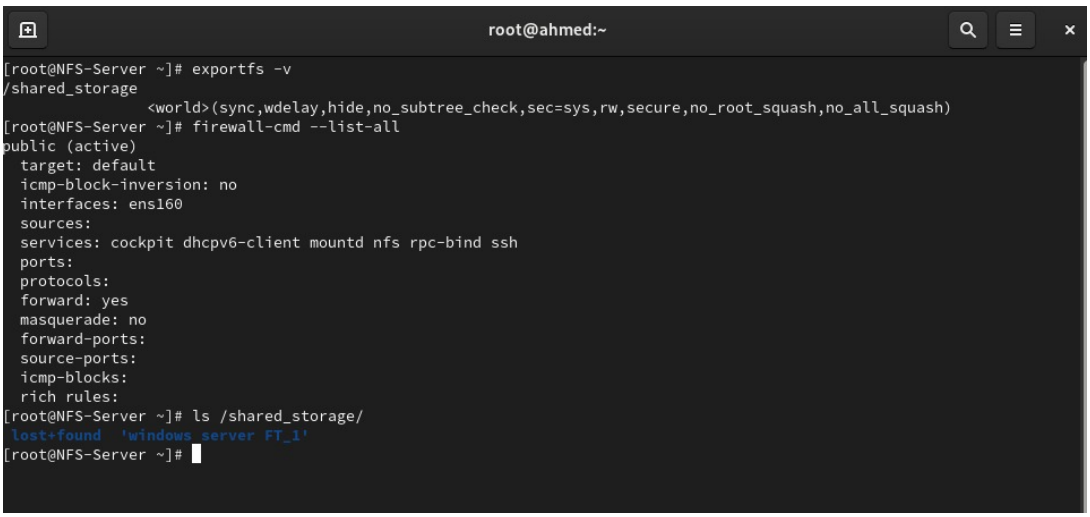

Figure 28: NFS export verification showing /shared_storage exported with sync, no_root_squash options, and firewall configured to allow NFS services on ens160 interface

5.2 Adding NFS Datastore to vCenter

After configuring the NFS server, the shared storage was added as a datastore in vCenter.

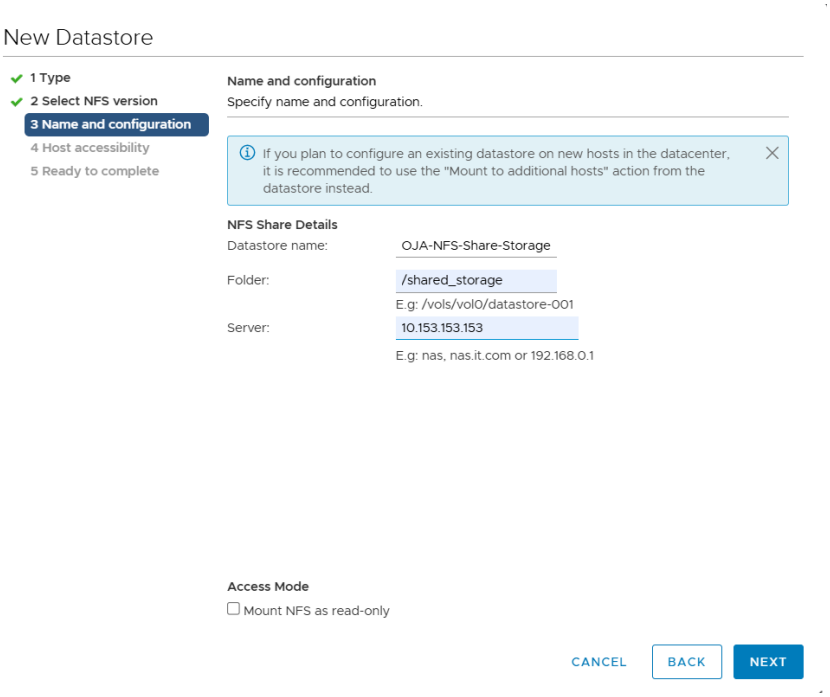

Figure 29: Creating NFS datastore "OJA-NFS-Share-Storage" pointing to /shared_storage folder on NFS server (10.153.153.153)

5.3 Virtual Machine Deployment on NFS Storage

To verify the NFS datastore functionality, a virtual machine was created using the shared storage.

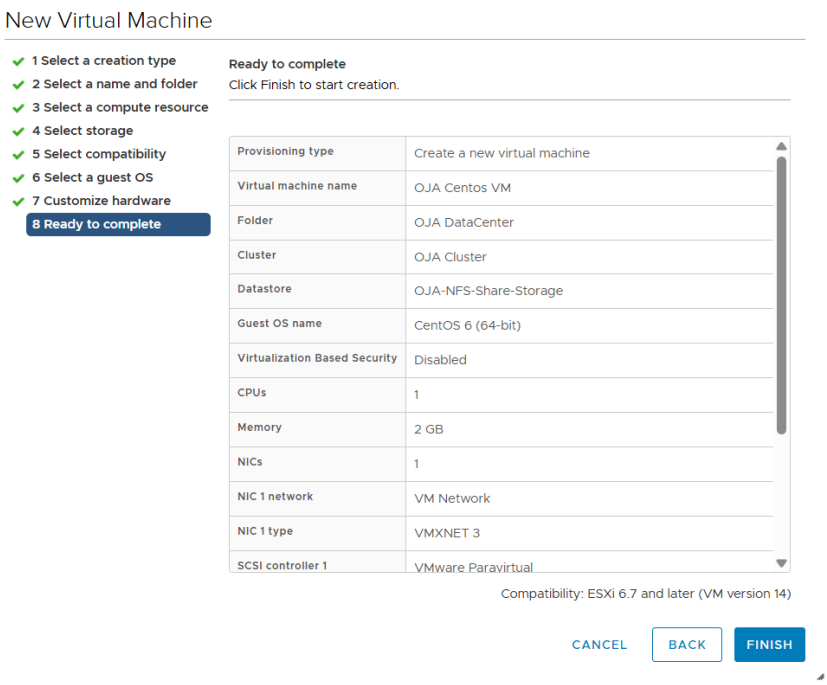

Figure 30: Creating "OJA Centos VM" virtual machine on OJA-NFS-Share-Storage datastore with CentOS 6 (64-bit), 1 CPU, 2 GB memory, and VM Network connectivity

5.4 NFS Storage Verification

```
[root@ahmed shared_storage]# ls
 dirfromme   lost+found  'OJA Centos VM'
[root@ahmed shared_storage]# ls 'OJA Centos VM'
'OJA Centos VM-a9bfb325.vswp'  'OJA Centos VM.vmx'
'OJA Centos VM-flat.vmdk'      'OJA Centos VM.vmx.lck'
'OJA Centos VM.nvram'           vmware.log
'OJA Centos VM.vmdk'           'vmx-OJA Centos VM-2847912741-1.vswp'
'OJA Centos VM.vmsd'
```

Figure 31: Linux NFS server /shared_storage directory showing OJA Centos VM files including .vmx, .vmdk, .nvram, .vmsd, and .vswp files, confirming successful VM storage on NFS share

6 Virtual Machine Operations

6.1 Cloning Virtual Machine to Template

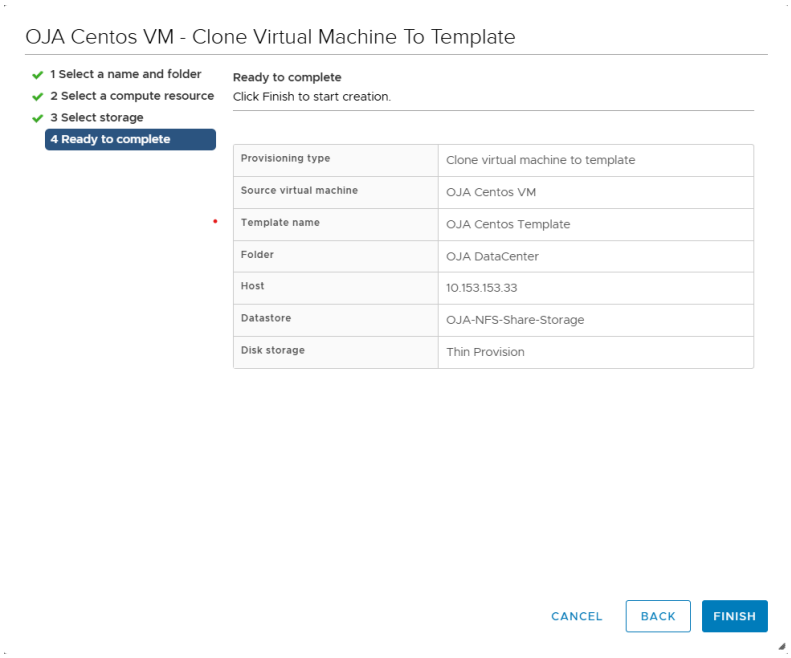

Figure 32: Cloning "OJA Centos VM" to template "OJA Centos Template" on OJA-NFS-Share-Storage datastore with Thin Provision disk format

6.2 Cloning Template to Content Library

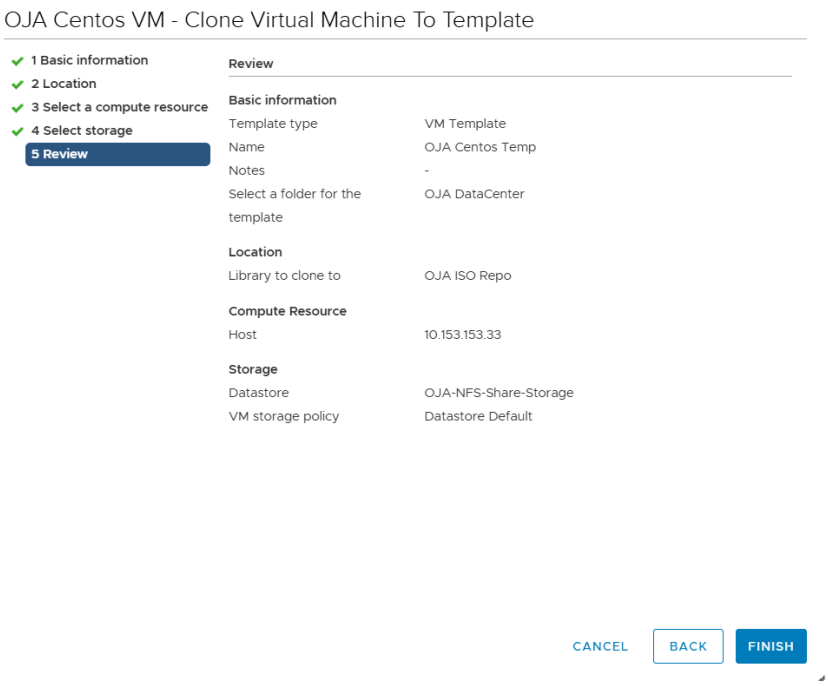

Figure 33: Cloning VM to template "OJA Centos Temp" in OJA ISO Repo content library, stored on OJA-NFS-Share-Storage

6.3 Cloning Existing Virtual Machine

VM cloning creates an identical copy of an existing virtual machine, useful for testing or scaling deployments.

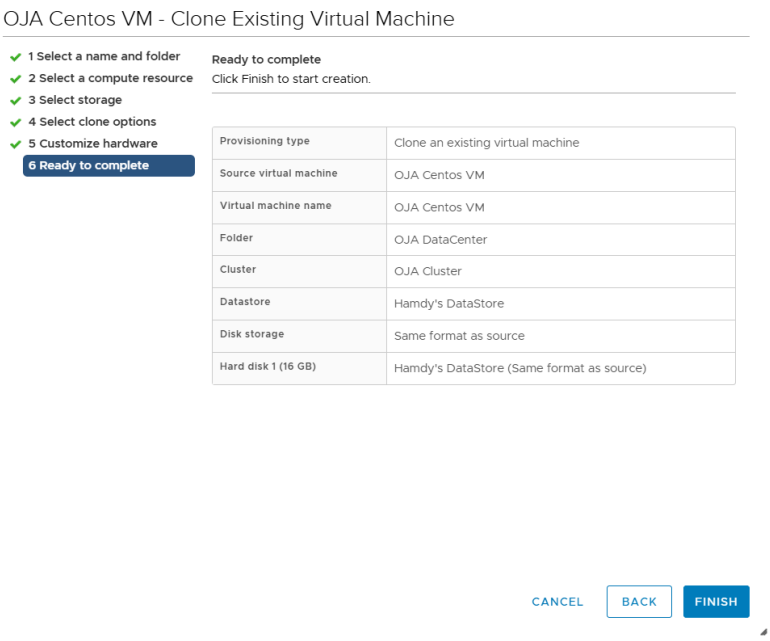

Figure 34: Cloning "OJA Centos VM" to create duplicate VM on Hamdy's DataStore in OJA Cluster, using same disk format as source (16 GB)

6.4 Creating VM Snapshots

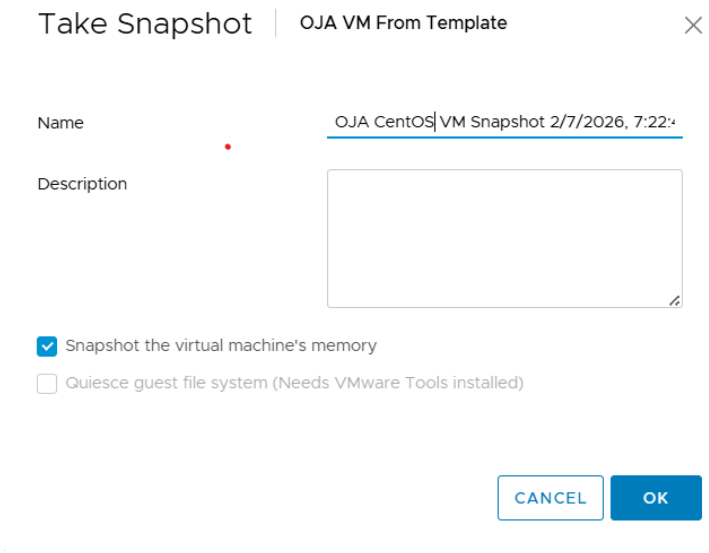

Figure 35: Creating snapshot "OJA CentOS VM Snapshot 2/7/2026" with memory state included for OJA VM From Template

6.5 Virtual Machine Migration (vMotion)

vMotion enables live migration of running virtual machines between ESXi hosts without downtime. Both compute and storage migration are supported.

6.5.1 Selecting Target ESXi Host

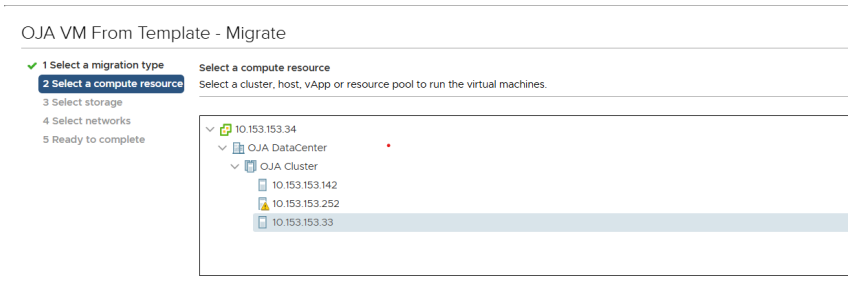

Figure 36: Selecting target ESXi host for VM migration within OJA Cluster for host 10.153.153.33 (ESXi-Hamdy)

6.5.2 Selecting Target Storage

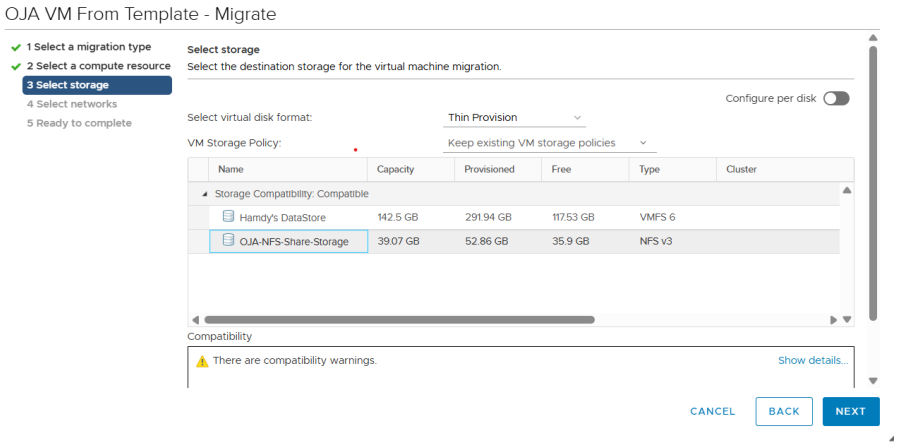

Figure 37: Selecting destination storage for VM migration - OJA-NFS-Share-Storage (NFS v3, 39.07 GB capacity, 35.9 GB free) with Thin Provision disk format

7 vSphere High Availability (HA) Configuration

7.1 Enabling vSphere HA

vSphere High Availability provides automatic restart of virtual machines in the event of host failure, ensuring business continuity with minimal downtime.

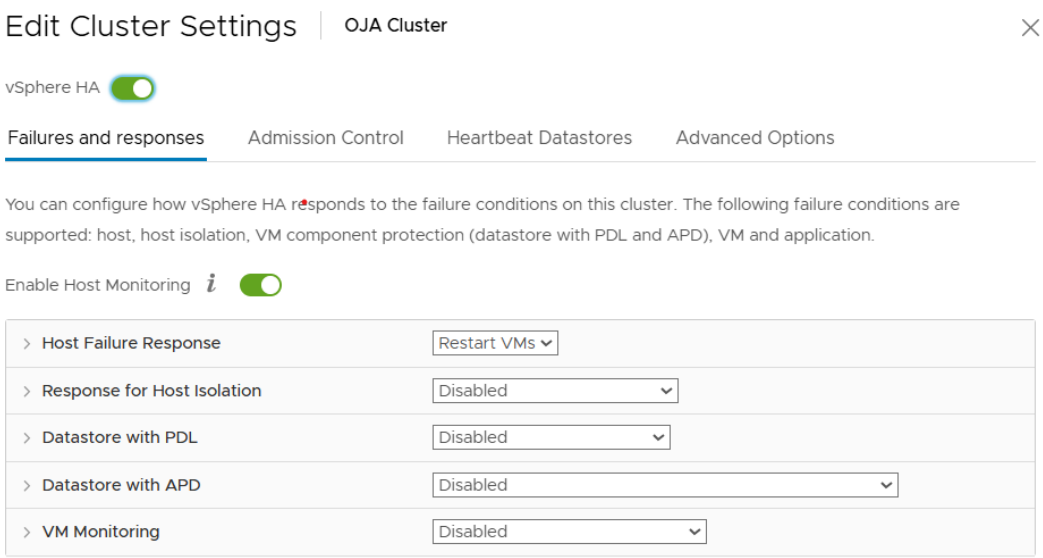

Figure 38: vSphere HA enabled on OJA Cluster with Host Monitoring enabled and Host Failure Response set to "Restart VMs"

7.2 HA Failover Verification

To verify HA functionality, a test scenario was executed by simulating host failure.

7.2.1 VM Before Host Failure

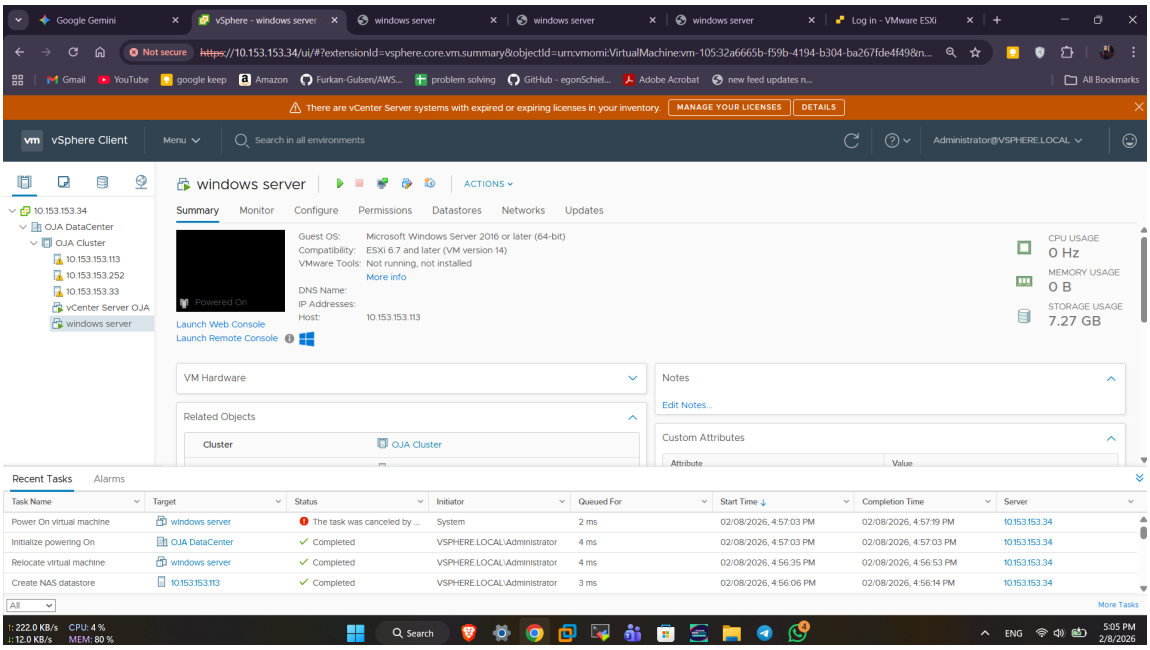

Figure 39: Windows Server VM running on ESXi host 10.153.153.113 (ESXi-Jimmy) before simulated host failure - Recent Tasks show normal operations

7.2.2 VM After Automatic Failover

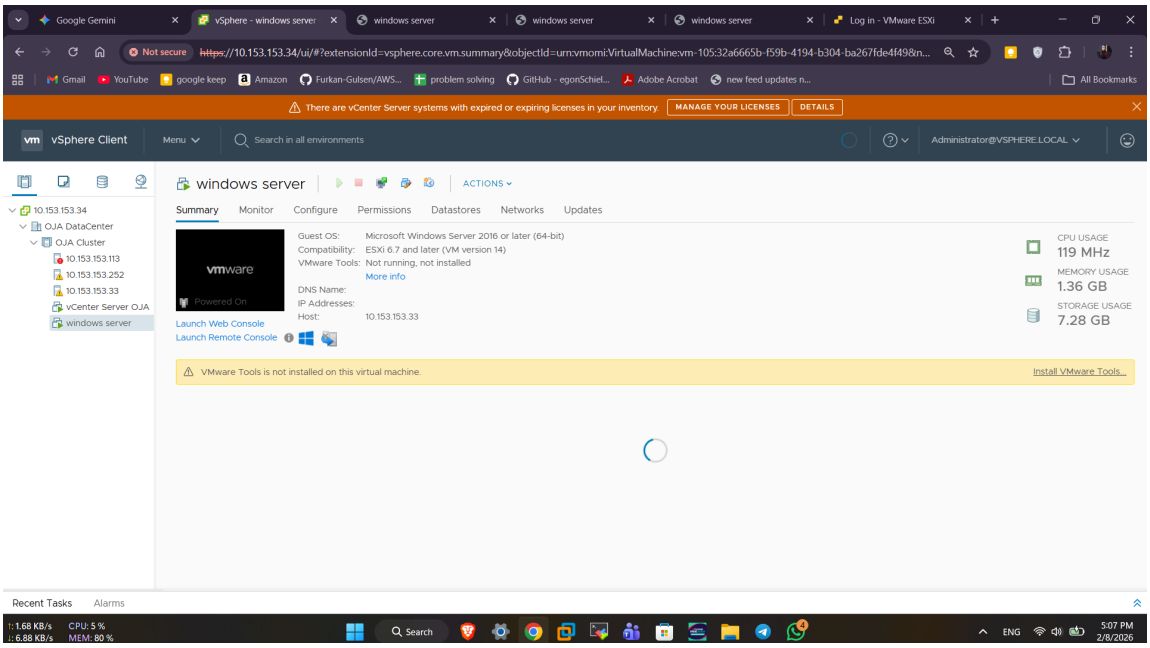

Figure 40: Windows Server VM automatically restarted on ESXi host 10.153.153.33 (ESXi-Hamdy) after HA-triggered failover, demonstrating successful automatic VM recovery

8 Distributed Resource Scheduler (DRS) Configuration

8.1 Enabling and Configuring DRS

DRS automatically balances computing workloads across ESXi hosts in a cluster by migrating virtual machines to optimize resource utilization.

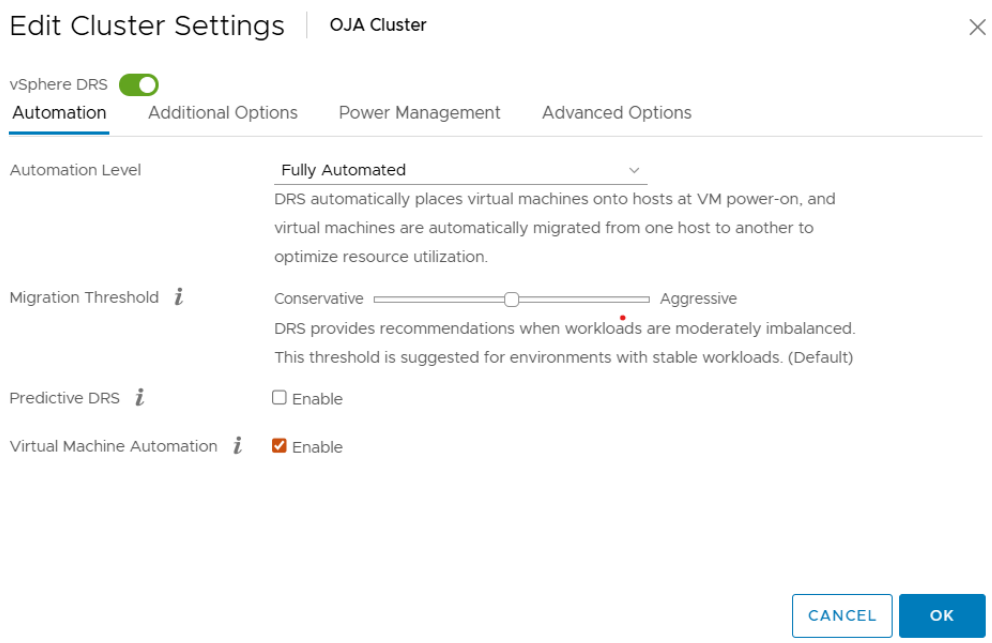

Figure 41: vSphere DRS enabled on OJA Cluster with Fully Automated mode, conservative-to-aggressive migration threshold, and Virtual Machine Automation enabled

8.2 DRS Load Balancing Verification

To verify DRS functionality, workload imbalance was created by running resource-intensive operations on one host.

8.2.1 VM Before DRS Migration

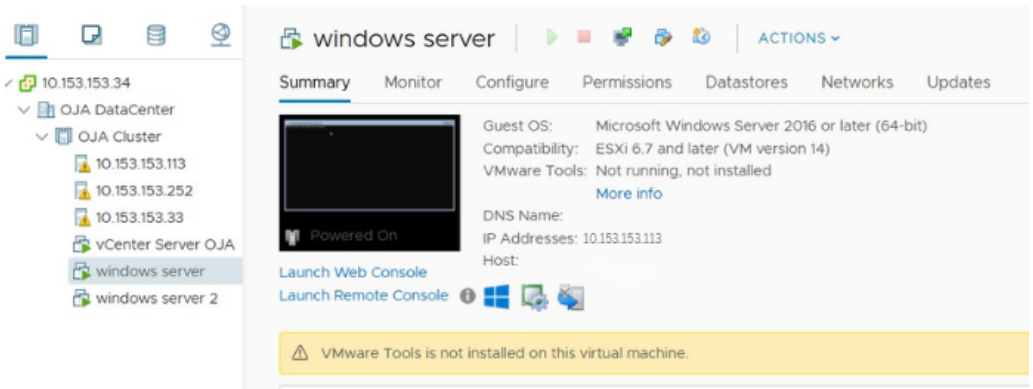

Figure 42: Windows Server VM initially running on ESXi host 10.153.153.113 (ESXi-Jimmy) before DRS-triggered migration

The Windows Server VM was originally running on ESXi-Jimmy (10.153.153.113).

8.2.2 VM After DRS Migration

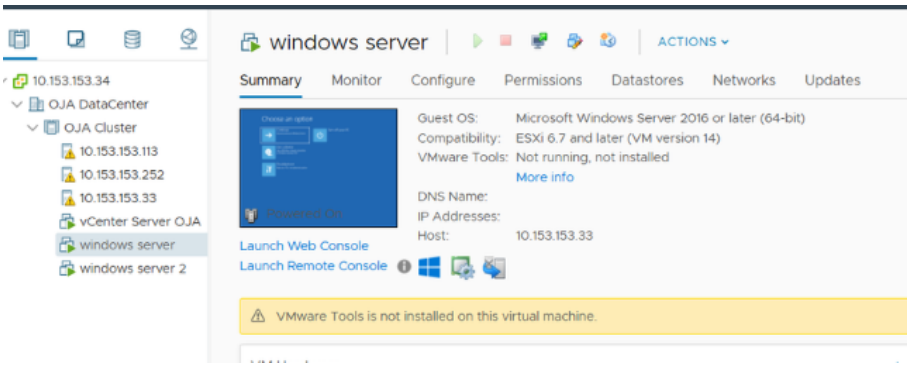

Figure 43: Windows Server VM automatically migrated via vMotion to ESXi host 10.153.153.33 (ESXi-Hamdy) after DRS detected resource imbalance

When heavy workload was detected on ESXi-Jimmy (10.153.153.113), DRS automatically triggered a live vMotion migration to balance resources across the cluster. The Windows Server VM was seamlessly migrated to ESXi-Hamdy (10.153.153.33) without downtime.

9 vSphere Fault Tolerance (FT) Configuration

9.1 Overview of Fault Tolerance

vSphere Fault Tolerance provides continuous availability by creating a live shadow instance of a virtual machine on another host. If the primary VM fails, the secondary VM takes over without zero downtime and no data loss.

9.2 Enabling Fault Tolerance Logging

Before enabling FT, the VMkernel port must be configured to support Fault Tolerance logging traffic.

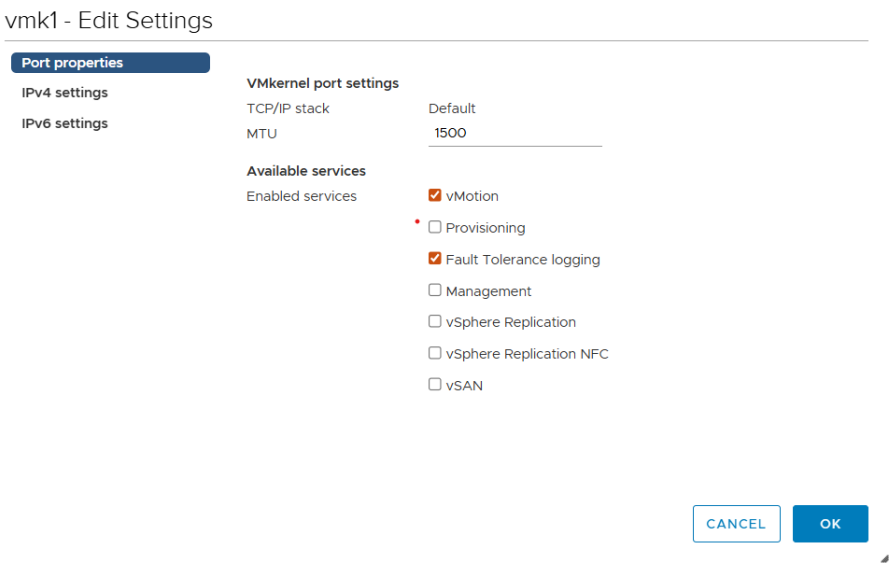

Figure 44: VMkernel port vmk1 configured with Fault Tolerance logging enabled alongside vMotion, using MTU 1500 and default TCP/IP stack

9.3 Configuring Fault Tolerance for Virtual Machine

9.3.1 Selecting Datastore for Secondary VM

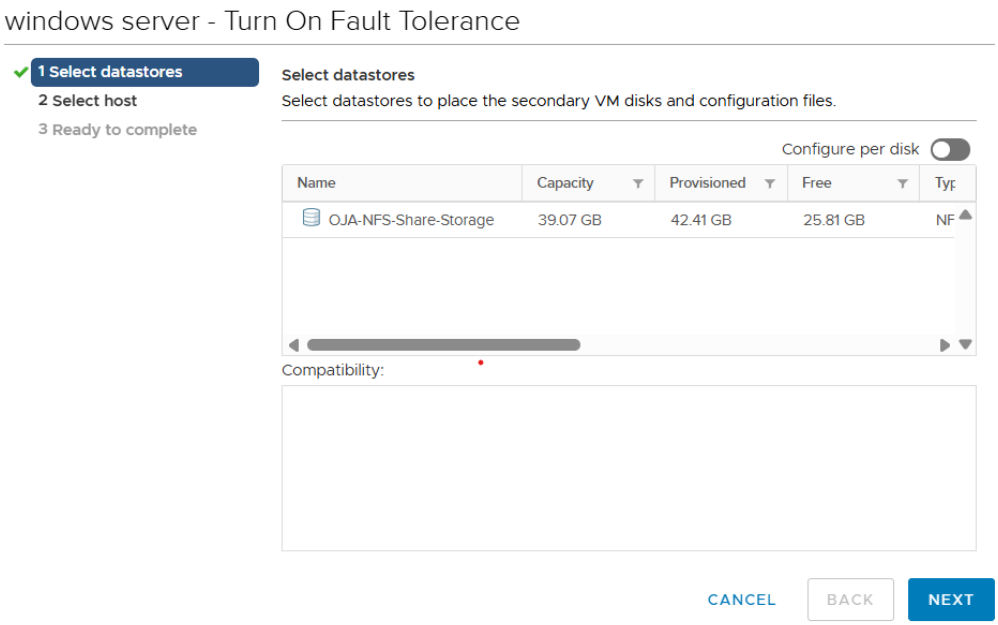

Figure 45: Selecting OJA-NFS-Share-Storage datastore (39.07 GB capacity, 25.81 GB free) for secondary VM disks and configuration files

9.3.2 Selecting Secondary Host

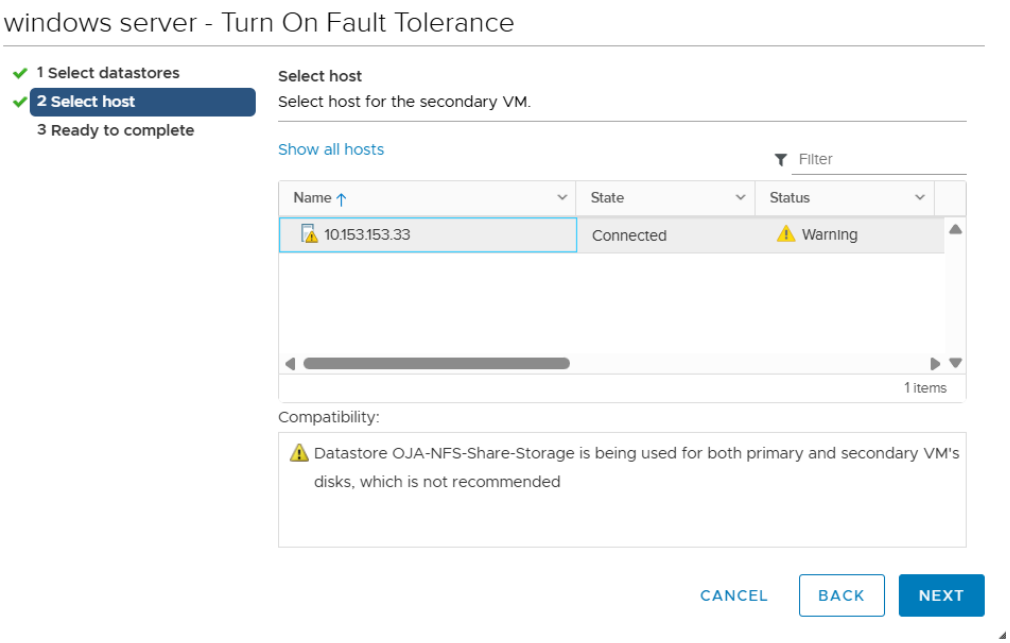

Figure 46: Selecting ESXi host 10.153.153.33 (ESXi-Hamdy) to run the secondary VM with warning about using same datastore for primary and secondary VMs

9.4 Fault Tolerance Verification

After enabling Fault Tolerance, verification was performed to ensure proper FT protection and failover capability.

9.4.1 FT Status Confirmation

Figure 47: Fault Tolerance status panel showing "Protected" state with secondary VM location on host 10.153.153.33 and running FT protection

9.4.2 Primary VM Running on ESXi-Jimmy

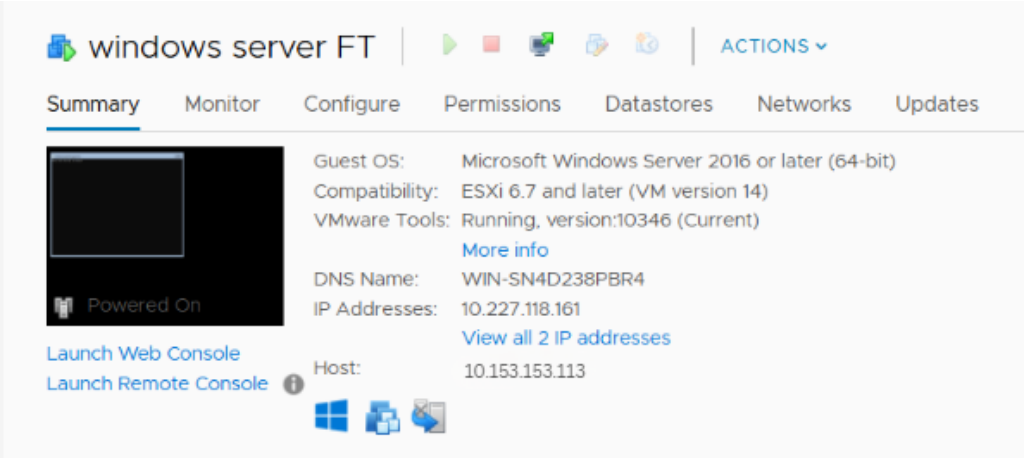

Figure 48: Windows Server FT VM running on ESXi host 10.153.153.113 (ESXi-Jimmy)

9.4.3 Seamless Failover Verification

Figure 49: Windows Server FT VM now running on ESXi host 10.153.153.33 (ESXi-Hamdy)

The verification demonstrates that Fault Tolerance successfully maintains continuous VM availability during host transitions. The VM seamlessly failed over from ESXi-Jimmy (10.153.153.113) to ESXi-Hamdy (10.153.153.33) while preserving its network identity (IP address and DNS name), confirming zero-downtime protection. The secondary VM automatically becomes the new primary when the original primary host experiences failure, and a new secondary VM is instantly created on another available host in the cluster.

10 Conclusion

This project successfully implemented a comprehensive VMware vSphere virtualization infrastructure with enterprise-grade features including Active Directory integration with the iti.local domain, vCenter Server management, three-node ESXi cluster, NFS shared storage, content library, and dedicated vMotion networking.
Advanced features successfully deployed and verified include vSphere High Availability (HA) for automatic VM failover, Distributed Resource Scheduler (DRS) for workload balancing, and Fault Tolerance (FT) for zero-downtime continuous availability. The implementation demonstrates a production-ready virtualization platform capable of supporting enterprise workloads with high availability and fault tolerance.

11 Acknowledgment

We express our sincere gratitude to **ENG. Ekram** for her exceptional guidance, continuous support, and expert supervision throughout this project. Her extensive knowledge and dedication were instrumental in our success and professional development.
Thank you for being an outstanding mentor.